

Adaptive Traffic Control System Implementation at HDBRTS Corridor

Srinivas Naik. Guguloth
ITNG, CDAC
Thiruvananthapuram, India
naik@cdac.in

Satheesh G
ITNG, CDAC
Thiruvananthapuram, India
satheeshg@cdac.in

Ravindra S Dhaded
HDBRTCo
Hubballi, India
rsdhaded@gmail.com

Benoygopal E B
ITNG, CDAC
Thiruvananthapuram, India
benoy@cdac.in

Dr. Prakash Rosayyan
ITNG, CDAC
Thiruvananthapuram, India
prakashr@cdac.in

Abhilash M
ITNG, CDAC
Thiruvananthapuram, India
mabhilash@cdac.in

Hemant Jeevan Magadam
ITNG, CDAC
Thiruvananthapuram, India
hemant@cdac.in

Abstract— The Adaptive Traffic Control System (ATCS) for the Hubballi-Dharwad Bus Rapid Transit System (HDBRTS-ATCS) corridor aims to optimize traffic flow and improve public transport efficiency in the region. This study evaluates the deployment and impact of the ATCS which adjusts traffic signals in the real-time based on the current traffic conditions. The implementation involved a combination of advanced sensor technologies, real-time data processing, and adaptive algorithms. Results indicate a significant improvement in traffic flow, reduced congestion and enhanced reliability of bus services. This paper highlights the effectiveness of the ATCS in managing traffic at a major BRT corridor and provides insights for similar implementation in other cities. **Reduce Travel Time: The travel time in the Hubballi-Dharwad ATCS corridor has been reduced from 55 minutes to 35 minutes along 22.5km dedicated corridor. Efficient Organ Transport Facilitated by Green Corridor Implementation in HDBRTS Corridor.**

The Composite Signal Control Strategy (CoSiCoSt) is an Adaptive Traffic Control System (ATCS) software algorithm for optimizing signal timings for the typical Indian driving and traffic conditions such as poor lane discipline and high heterogeneity. Achieving Green Signals at Intersection with CoSiCoSt algorithms, the average travel time was reduced from 4.43 minutes to 2.21 minutes. With the implementation of TSP, the average delay for clearance of BRT buses reduced from 17 seconds to 7 seconds.

Keywords— *BRTS, Adaptive Traffic Control System, ATCS, Traffic Controller, TraMM-EnV, CoSiCoSt, CUTE and TSP.*

I. INTRODUCTION

As part of the implementation of the project, traffic signal and pedestrian controllers have been installed in junctions inside and outside the dedicated corridor where the BRT buses would operate for the smooth and safe operation of motor vehicles and pedestrians crossing the motorways. Traffic signal installations completed at 32 and 02 Pedestrian Midblock Crossings in Hubballi - Dharwad Bus Rapid Transit System Corridor (HDBRTS), Junction Signal and Midblock are connected to Adaptive Traffic Control System (ATCS) and are monitored from there.

The travel time in the Hubballi-Dharwad ATCS corridor has been reduced from 55 minutes to 35 minutes along 22.5km dedicated corridor. Efficient Organ Transport Facilitated by Green Corridor Implementation in HDBRTS Corridor. Achieving Green Signals at Intersection with CoSiCoSt

algorithms, the average travel time was reduced from 4.43 minutes to 2.21 minutes.

With the implementation of Transit Signal Priority (TSP). TSP is to reduce the time transit vehicles stopped at red lights, thus improving service reliability, reducing overall travel time, and increasing the efficiency of public transportation system, the average delay for clearance of BRT buses reduced from 17 seconds to 7 seconds.

II. LITERATURE REVIEW

Paper [1] describes the Bus Rapid Transit System (BRT) traffic control system which uses fixed-cycle traffic lights. This results travel time improvement by 23 % to 33%. Paper [2] describes ACP based management and control system. Also it has feature of incident management, forecasting, and scheduling in real time. Paper [3] describes BRT control system without utilizing exclusive bus line results reduction in travel time. Paper [4] describes effective TSP method which reduces the average delay time to 10%. Paper [5] describes transit signal priority (TSP) based Bus Rapid Transit System (BRT) which operate on fixed-time signal schedules, adjusting traffic signals based on predefined timetables. It reduces the bus transit time to 37.40% compared to other TSP methods. Paper [6] describes adaptive Traffic Control System (ATCS) which adapts the timing of traffic signals based on the real demand using sensor. ATCS helped to reduce fuel consumption and air pollution. Paper [7] describes data-driven adaptive coordinated control algorithm for urban traffic which solve congestion problems and green time loss. Paper [8] describes adaptive timer control system which is based on the number of vehicles at the previous junction as well as the count of vehicles currently arriving at the intersections to update the signal timers. Also, it prevents delays in emergency response timers and provides a better impact on public safety and health outcomes as an added feature to prioritize emergency vehicles. Paper [9] describes Neural Network algorithm-based traffic control system improved traffic efficiency by adjusting green timing. This reduces average delay to 8.45%. Paper [10] describes Vehicular Adhoc Network based adaptive control system which is based on clustering. Paper [11] describes ATCS system for urban environment which adjusts signal timings based on input demand from the sensor. Paper [12] describes camera based adaptive traffic control

system which changes green timing based on camera sensor input. Paper [13] describes the adaptive control with data driven control model for urban traffic using macroscopic fundamental diagram. Paper [14] describes adaptive traffic regulation system with GSM based emergency preemption system. It improves city traffic control and regulation. This system gives priority to emergency vehicles to serve immediately. It is very important for life saving scenarios. Paper [15] describes the adaptive traffic control system which realize the fixed timing function in mixed traffic condition. It reduces the average driving 20%. Paper [16] describes smart Traffic signal system which checks traffic congestion. This improves traffic movement by using a data collection in real-time.

So the efficient implementation of indigenous adaptive traffic control system with its assessment result at heterogeneous Indian road condition which is described in this paper.

III. OBJECTIVE OF IMPLEMENTATION OF ATCS

A. Optimize Traffic Flow

To enhance overall traffic movement within the HDBRTS corridor by minimizing delays and congestion through real-time signal adjustments.

B. Improve Bus Service Efficiency

To reduce travel time and increase reliability of the BRT system by ensuring priority for buses and minimizing delays at intersections.

C. Reduce Environments Impact

To decrease vehicle emissions and fuel consumption by improving traffic flow and reducing idle times at traffic signals.

D. Enhance Safety

To improve road safety by managing traffic more effectively and reducing the likelihood of accidents due to congestion and signal inefficiencies.

IV. SUB-SYSTEMS OF ATCS

The sub-systems of ATCS are CUTE Controller, TraMM-EnV Software and CoSiCoSt-EnV.

A. CUTE Controller

It is a vehicle actuated road traffic signal controller which operates as an isolated intersections, part of coordinated signal control system or as a part of ATCS. CUTE monitored, managed and controlled from the City Mobility Centre (CMC) over wired or wireless communication network like fiber optics, lease line or 4G/ 5G. The fig 1 shows picture of CUTE.



Fig. 1. CUTE controller cabinet.

B. Traffic Monitoring and Management Enhanced Version (TraMM-EnV) Software

TraMM-EnV is a software utility for monitoring and managing CUTE traffic junction controllers CUTE remotely from a Central Server. TraMM-EnV displays status of traffic signal in Human Machine Interface- HMI and reports. It also provides traffic plan configuration tool with upload and download feature, remote administration of junction from central control room. TraMM-EnV dashboard provides GIS enabled monitoring and control feature of traffic signal. The dashboard snapshot shown in fig 2.

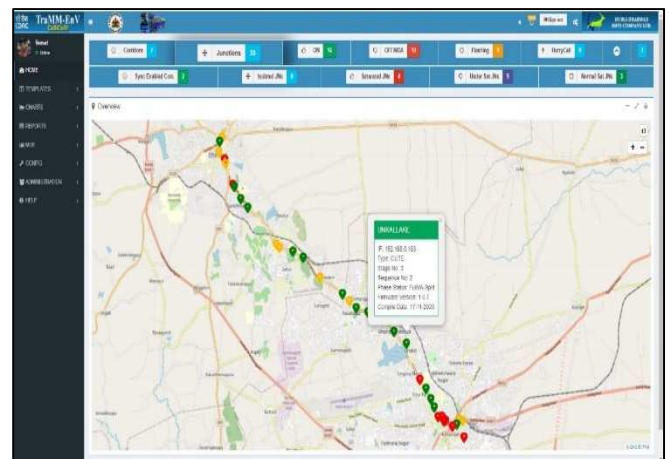


Fig. 2. TraMM-EnV dashboard

C. C-DAC CoSiCoSt (Composite Signal Control Strategy)

The ATCS in Hubballi -Dharwad BRTS is designed to enhance traffic management to reduce stops and delays and ensure smooth and efficient movement of BRT buses along the corridor. The key components and functionalities of ATCS in this BRT system are as follows.

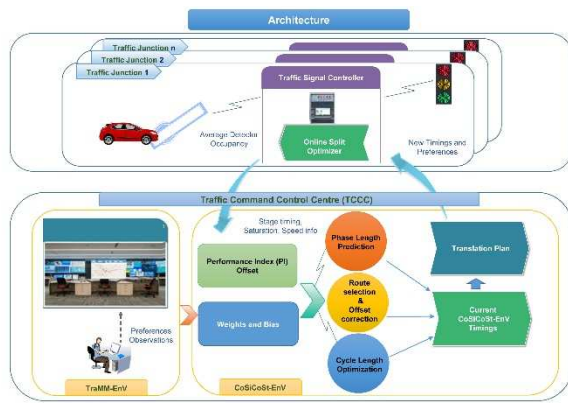


Fig. 3. CoSiCoSt Architecture

a) *Real-Time Traffic Monitoring*: ATCS employs sensors - cameras installed at key junctions and intersections to monitor real-time traffic conditions. These device collect data on vehicle flow, speed, congestion levels, and traffic patterns.

b) *Centralized Traffic Management and control*: A centralized traffic management centre oversees the entire ATCS. ATCS adjusts timings of traffic signal based on presence of vehicle and reduces travel time. Signal Priority for BRT Buses prioritizes BRT buses at intersections, ensuring minimal waiting times and faster transit through the corridor.

c) *Traffic Signal Coordination*: Traffic signals along the BRT corridor are synchronized to create green waves for BRT buses, minimizing stops and ensuring a smooth journey.

Inter-Junction Coordination is the Coordination between different junctions which ensures that changes in one junction's signal timing do not adversely affect traffic flow at adjacent junctions.

The aim of CoSiCoSt is to reduce the stops and traffic delays in a in Indian mixed traffic conditions. It operates in real time by calculating the optimal cycle times and optimizing split and offset. These calculations will be based on data timing which varied based on vehicle detection sensors.

Working methodology steps of CoSiCoSt

1. Identify a critical junction
2. Determine a common cycle time
3. Set the upper and lower Degree of Saturation (DOS) thresholds
4. Monitor the Degree of Saturation (DOS)
5. Calibrate the stage timings
6. Establish a priority route
7. Calculate the offset for each junction (Validate the speeds and offsets)
8. Adjust the priority stage
9. Propose new Cycle length and phase timings based on DOS
10. Fallback mode: In the event of a network interruption, the fallback mode is the Full Vehicle Actuation split

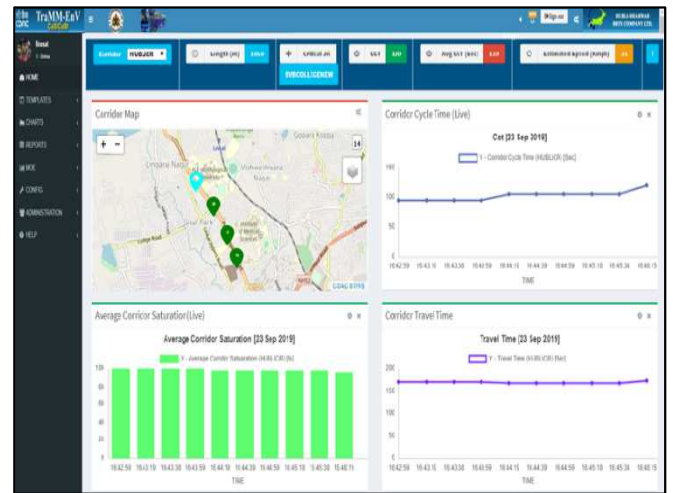


Fig. 4. ATCS Corridor level Traffic Signal Coordination

The ATCS system has the facility to Traffic signal coordination in the corridor level and optimizes the performance parameters such as speed, travel time, speed, congestion levels and improving overall efficiency.

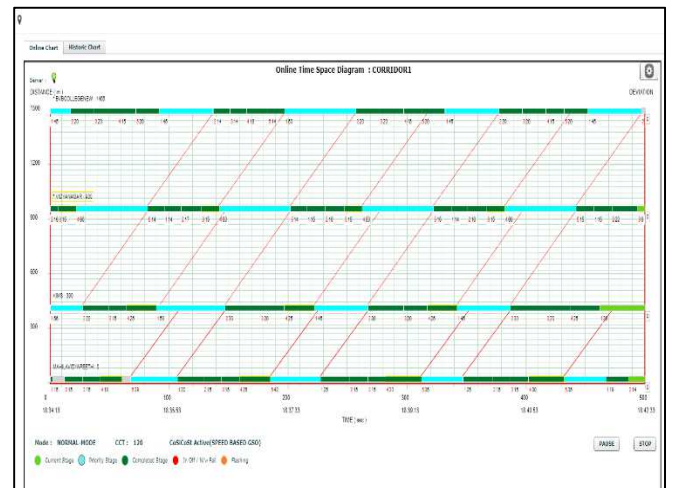


Fig. 5. Priority stage (indicated in blue color) shows the synchronization band between MVP to BVB Junction at Hubballi HDBRTS.

V. PRIORITY CLEARANCE FOR BRT BUSES IN HUBBALLI-DHARWAD CORRIDOR

The introduction of Bus Rapid Transit (BRT) buses in Hubballi-Dharwad corridor as shown significant improvements in travel time efficiency. Studies and practical implementation have demonstrated that Transit Signal Priority (TSP) can effectively reduce delays and improve the punctuality of BRT services. Real time TSP works by adjusting timings of traffic signals to prioritize the passage of BRT buses, thereby minimizing stops at intersections and reducing overall travel time.

VI. IMPLEMENTATION OF TRANSIT SIGNAL PRIORITY (TSP) IN HUBBALLI- DHARWAD CORRIDOR

Without Transit Signal Priority (TSP), average time delay for clearance of BRT buses was 17 seconds. After successful implementation of TSP, the average time delay was 7 seconds.

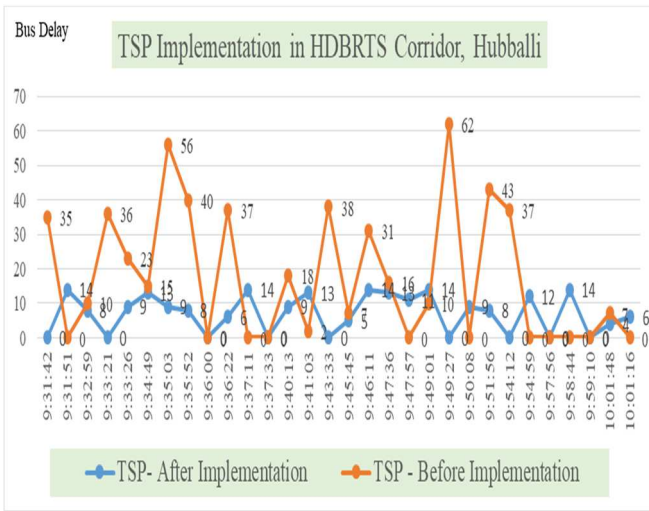


Fig. 6. TSP- After and Before Implementation

The immediate clearance time for Transit Signal Priority – TSP in the Hubballi-Dharwad BRTS corridor in Karnataka is typically designed to minimize delays for BRT buses by giving them priority at intersections. The TSP system reduces bus travel time by granting signal priority, thus decreasing the waiting time at traffic signals. This mechanism is part of the broader Intelligent Transportation System (ITS) aimed at improving public transit efficiency

VII. BENEFITS OF ATCS IMPLEMENTATION IN HUBBALLI-DHARWAD BRTS CORRIDOR

The benefits are given below.

a) *Reduced Travel Time*: The dynamic adjustment of signal timings and priority for BRT buses significantly reduces travel time for passengers. Specifically, the travel time in the Hubballi-Dharwad ATCS corridor has been reduced from 55 minutes to 35 minutes along 22.5km dedicated corridor.

b) *Efficient Organ Transport Facilitated by Green Corridor Implementation in HDBRTS Corridor*: Several green corridor were established from SDM Hospital, Dharwad to Narendra Cross due to brain stem death cases at SDM Hospital. To facilitate the transport of organs, a green corridor was arranged through the HDBRTS Corridor. The successful creation of these green corridors was managed by the Adaptive Traffic Control System – ATCS.

c) *Achieving Green Signals at Intersections with CoSiCoSt Algorithms*: Implemented Composite Signal Control Strategy (CoSiCoSt) algorithms to achieve green signals at traffic intersections from MVP junction to BVB junction. The results demonstrated that all the intersections were able to provide green signals effectively. As a result, average travel time was reduced from 4.43 minutes to 2.21 minutes with the implementation of CoSiCoSt algorithms as shown below table I, II, III and IV.

TABLE I. GREEN SIGNALS WITH CoSiCoSt AT 9.00 TO 10.00

Without CoSiCoSt					
BRT Bus delay report- Time – 9.0.0 to 10.0.0					
Sl. No.	Bus Arrival Time	Signal Status (Put Tick mark in any one of the following)			Travel time (in seconds)
		Green	Amber	Red	
1	9:00:30	✓			0
2	9:05:10			✓	10
3	9:13:00		✓		0
4	9:25:00			✓	20
Average travel time in seconds					7.5
With CoSiCoSt					
BRT Bus delay report- (Mahila Vidyapeeth to BVB college)					
Time – 9.00 to 10:00					
Sl. No .	Travel Start time		Travel end time		Travel time (in minutes)
1	9:00:30		9:30:20		02:50
2	9:05:20		9:08:30		03:10
3	9:30:00		9:31:30		01:30
4	9:45:30		9:47:00		01:30
Average travel time from MVP to BVB in minutes					02:15

TABLE II. GREEN SIGNALS WITH CoSiCoSt AT 13.00 TO 14.00

Without CoSiCoSt					
BRT Bus delay report- Time – 13.0.0 to 14.0.0					
Sl. No.	Bus Arrival Time	Signal Status (Put Tick mark in any one of the following)			Travel time (in seconds)
		Green	Amber	Red	
1	13:00:30	✓			0
2	13:05:30		✓		0
3	13:09:20	✓			0
4	13:40:30			✓	20
Average travel time in seconds					5
With CoSiCoSt					
BRT Bus delay report- (Mahila Vidyapeeth to BVB college)					
Time – 13.00 to 14:00					
Sl. No	Travel Start time		Travel end time		Travel time (in minutes)
1	13:00:30		13:02:30		02:00
2	13:20:30		13:23:10		02:40
3	13:40:00		13:42:50		02:50
4	13:50:30		13:51:50		01:20
Average travel time from MVP to BVB in minutes					02:12

TABLE III. GREEN SIGNALS WITH CoSiCoSt AT 18.00 TO 19.00

Without CoSiCoSt					
BRT Bus delay report- Time – 18.0.0 to 19.0.0					
Sl. No.	Bus Arrival Time	Signal Status (Put Tick mark in any one of the following)			Travel time (in seconds)
		Green	Amber	Red	
1	18:00:30	✓			0
2	18:08:10			✓	30
3	18:20:30		✓		0
4	18:30:00			✓	5
Average travel time in seconds					8.75
With CoSiCoSt					
BRT Bus delay report- (Mahila Vidyapeeth to BVB college)					
Time – 18.00 to 19:00					
Sl. No	Travel Start time		Travel end time		Travel time (in minutes)
1	18:00:30		18:02:20		01:50
2	18:20:30		18:23:30		03:00
3	18:40:00		18:42:40		02:40
4	18:50:30		18:53:20		02:50
Average travel time from MVP to BVB in minutes					02:35

TABLE IV. GREEN SIGNALS WITHOUT CoSiCoSt AT 9.00 TO 10.35

Without CoSiCoSt BRT Bus travel time report- (Mahila Vidyapeeth to BVB college) Time – 9.00 to 10.35 A.M.				
Sl. No.	Travel Start time	Travel end time	Travel time (in minutes)	Remark/ Status
1	9:10:50	9:14:34	03:44	Limited Stops
2	9:23:44	9:28:39	04:55	MVP to BVB
3	9:31:36	9:35:44	04:08	BVB to MVP
4	9:37:08	9:41:31	04:23	MVP to BVB
5	9:45:52	9:53:54	08:02	BVB to MVP
6	9:57:02	10:01:02	04:00	MVP to BVB
7	10:05:31	10:09:19	03:48	BVB to MVP
8	10:16:29	10:21:48	05:21	MVP to BVB
9	10:27:10	10:31:15	04:05	BVB to MVP
Average travel time from MVP to BVB in minutes			04:43	

VIII. ADVANTAGES AND BENEFITS OF CoSiCoSt IN HUBBALLI- DHARWAD CORRIDOR

Following are the advantages which achieved after implementing CoSiCoSt in Hubballi- Dharwad Corridor.

a) Improved Service Reliability: Real time monitoring and adjustments enhance the reliability and punctuality of the BRT service.

b) Decreased Congestion: Optimized traffic flow leads to reduced congestion along the BRT corridor and adjacent roads.

c) Enhanced Safety: Better traffic management and reduced congestion contribute to improved road safety for all users.

d) Environmental Impact: Smoother traffic flow and reduced congestion result in lower vehicle emissions, contributing to a cleaner environment.

Following are the benefits which achieved after implementing CoSiCoSt in Hubballi- Dharwad Corridor

a) The project has helped in growth of civic sense in the citizens

b) The project has ensured the safety of both riders & pedestrians.

c) The project resulted in smooth flow of traffic in the twin cities of Hubballi-Dharwad.

d) The project has reduced delay in crossing the junctions.

e) The project has ensured the shorter queue lengths at signal crossings.

f) The project has decreased travel time of commuters.

IX. CHALLENGES AND CONSIDERATIONS

Following are the challenges faced during implementation of CoSiCoSt in Hubballi- Dharwad Corridor.

a) Initial Investment: Implementing ATCS involves substantial initial investment in infrastructure, technology, and training.

b) Maintenance and Upkeep: Continuous maintenance of sensors - cameras, and control systems is crucial for the sustained success of ATCS.

c) Public Awareness: Educating the public and ensuring their cooperation is vital for the smooth functioning of the ATCS and overall BRT system.

X. CONCLUSION

With Implementation of Composite Signal Control Strategy (CoSiCoSt) algorithms to achieve green signals at traffic intersections (from MVP to BVB college junctions). The results demonstrated that all the intersections were able to provide green signals effectively. As a result, average travel time was reduced from 4.43 minutes to 2.21 minutes with the implementation of CoSiCoSt algorithm.

Several green corridor were established from SDM Hospital, Dharwad to Narendra Cross due to brain stem death cases at SDM Hospital. To facilitate the transport of organs, a green corridor was arranged through the HDBRTS Corridor. The successful creation of these green corridors was managed by the ATCS.

The dynamic adjustment of signal timings and priority for BRT buses significantly reduces travel time for passengers. Specifically, the travel time in the Hubballi-Dharwad ATCS corridor has been reduced from 55 minutes to 35 minutes along 22.5km dedicated corridor.

Prior to the implementation of Transit Signal Priority (TSP), the average time delay for clearance of BRT buses was 17 seconds. After successful implementation of TSP, the average time delay was 7seconds.

The power consumption for a typical 4-arm junction is 220 watts, with a current of 8.5A at 24 volts for the Traffic Signal Controller.

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